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Question 1 [C01] [10 Points]

Write a java program to solve the following. A hotel has two room types: Studio and Apartment. The price per night depends on the month number (1 = January, 5 = May, etc.)

Month Number	Month Name	Studio(USD)	Apartment(USD)
5	May	50	65
6	June	75	69
7	July	76	77
8	August	76	77
9	September	75	69
10	October	50	65

Only these 6 months are valid. No need to handle any other month number.
If more than 14 nights:

Month(s)	Studio Discount	Apartment Discount
May or October (5/10)	30% off	10% off
June or September (6/9)	20% off	10% off
July or August (7/8)	No discount	10% off

Input	Output
Enter month: 5 Enter Total Nights: 15	Apartment: 877.5 USD. Studio: 525.0 USD.
Enter month: 8 Enter Total Nights: 13	Apartment: 1001.0 USD. Studio: 988.0 USD.

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Question 1 [C01] [10 Points]

Write a java program to solve the following. A beach resort offers Bungalow and Villa rooms. Price per night depends on month number (5 = May, ..., 10 = October):

Month	Bungalow (USD)	Villa (USD)
5 (May)	55	80
6 (June)	85	120
7 (July)	95	135
8 (August)	95	135
9 (September)	85	120
10 (October)	55	80

Only these 6 months are valid. No need to handle any other month number.
Extra Charge Rule (only if nights > 12)

Month(s)	Bungalow Extra	Villa Extra
5 or 10 (May/Oct)	+5 USD/night	+8 USD/night
6 or 9 (Jun/Sep)	+3 USD/night	+8 USD/night
7 or 8 (Jul/Aug)	No extra	+8 USD/night

Extra charge is added per night after base price.

Input	Output
Enter month: 5 Enter nights: 15	Villa: 1320.0 USD. Bungalow: 900.0 USD.
Enter month: 7 Enter nights: 10	Villa: 1350.0 USD. Bungalow: 950.0 USD.

[SET-A]

SOLUTION

```
import java.util.Scanner;
public class HotelPricing {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter month: ");
        int month = sc.nextInt();

        System.out.print("Enter Total Nights: ");
        int nights = sc.nextInt();

        // ---- Base nightly prices ----
        double studioNight = 0.0;
        double apartmentNight = 0.0;

        if (month == 5 || month == 10) {      // May or October
            studioNight = 50.0;
            apartmentNight = 65.0;
        } else if (month == 6 || month == 9) { // June or September
            studioNight = 75.0;
            apartmentNight = 69.0;
        } else if (month == 7 || month == 8) { // July or August
            studioNight = 76.0;
            apartmentNight = 77.0;
        }

        // ---- Full price ----
        double totalStudio = studioNight * nights;
        double totalApartment = apartmentNight * nights;

        // ---- Discounts (only when nights > 14) ----
        if (nights > 14) {
            if (month == 5 || month == 10) {
                totalStudio *= 0.70; // 30 % off
                totalApartment *= 0.90; // 10 % off
            } else if (month == 6 || month == 9) {
                totalStudio *= 0.80; // 20 % off
                totalApartment *= 0.90; // 10 % off
            } else if (month == 7 || month == 8) {
                totalApartment *= 0.90; // 10 % off only
            }
        }

        System.out.println("Apartment: " + totalApartment + " USD.");
        System.out.println("Studio: " + totalStudio + " USD.");

        sc.close();
    }
}
```

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RUBRIC

SL	Points To Meet	Marks [10]
1	Exact prompts + correct Scanner input	1
2	All 6 months have correct prices via if-else	3
3	Correct % applied only when nights > 14	5
4	Correct Output	1

[SET-B]

SOLUTION

```
import java.util.Scanner;
public class BeachResort {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter month: ");
        int month = sc.nextInt();

        System.out.print("Enter nights: ");
        int nights = sc.nextInt();

        double bungalowBase = 0.0;
        double villaBase = 0.0;

        // Set base prices using if-else
        if (month == 5 || month == 10) {
            bungalowBase = 55.0;
            villaBase = 80.0;
        } else if (month == 6 || month == 9) {
            bungalowBase = 85.0;
            villaBase = 120.0;
        } else if (month == 7 || month == 8) {
            bungalowBase = 95.0;
            villaBase = 135.0;
        }

        // Calculate base total
        double totalBungalow = bungalowBase * nights;
        double totalVilla = villaBase * nights;

        // Apply extra charge only if nights > 12
        if (nights > 12) {
            double bungalowExtra = 0.0;
            double villaExtra = 0.0;

            if (month == 5 || month == 10) {
                bungalowExtra = 5.0;
                villaExtra = 8.0;
            } else if (month == 6 || month == 9) {
                bungalowExtra = 3.0;
                villaExtra = 8.0;
            } else if (month == 7 || month == 8) {
                villaExtra = 8.0;
            }

            totalBungalow += bungalowExtra * nights;
            totalVilla += villaExtra * nights;
        }
    }
}
```

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```
}

// Output: Villa first, then Bungalow
System.out.println("Villa: " + totalVilla + " USD.");
System.out.println("Bungalow: " + totalBungalow + " USD.");

sc.close();
}
```

RUBRIC

SL	Points To Meet	Marks [10]
1	Exact prompts + correct Scanner input	1
2	All 6 months have correct prices via if-else	3
3	Correct extra per night only when nights > 12, per month group	5
4	Correct Output	1